Comprehensive Guide to Autonomous AI Agents and RAG Systems

Chapter 1: Foundations of Retrieval-Augmented Generation

Retrieval-Augmented Generation (RAG) is an architectural pattern that combines parametric memory (pre-trained LLMs) with non-parametric retrieval memory (vector databases). By dynamically retrieving relevant document chunks from external stores, RAG reduces hallucinations and enables grounded answers.

Chapter 2: Vector Search with FAISS

FAISS (Facebook AI Similarity Search) is an open-source library developed by Meta for rapid similarity search of dense numerical vectors. In RAG pipelines, text chunks are embedded into high-dimensional vector spaces (such as 1536-dimensional embeddings produced by text-embedding-3-small). FAISS constructs index structures such as IndexFlatL2 or HNSW to execute nearest neighbor search in sub-millisecond latencies.

Chapter 3: Chunking Strategies

Document chunking splits lengthy texts into smaller, semantically coherent segments. A common configuration is a chunk size of 800 to 1200 characters with an overlap of 150 to 250 characters. The overlap prevents critical semantic context from being cleaved across chunk boundaries.